

SEIZURE PREDICTION USING LOGISTIC REGRESSION AND REGULARIZATION STRATEGIES

Effect of Preprocessing Choices, Model Complexity, and Regularization Strategies on
Generalisation Performance in Epileptic Seizure Prediction Tasks

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Abstract

This paper presents a comprehensive empirical investigation into how preprocessing pipeline design, regularization strategy, model complexity, and class imbalance handling collectively influence the generalisation performance of logistic regression classifiers in epileptic seizure prediction. Three EEG-based datasets of varying size, class imbalance severity, and feature representation were utilized: the UCI Epileptic Seizure Recognition dataset (11,500 samples, 178 features, 20% seizure prevalence), the Bonn University EEG dataset (500 samples, 12 statistical features, 20% seizure prevalence), and a Kaggle-style dataset (2,000 samples, 50 features, 10% seizure prevalence representing severe imbalance).

Two preprocessing pipelines were systematically compared: Pipeline A (StandardScaler normalization followed by ANOVA F-test feature selection) and Pipeline B (StandardScaler followed by Principal Component Analysis). Experiments demonstrate that preprocessing step ordering significantly affects model performance, with Pipeline A consistently outperforming Pipeline B by 0.023 to 0.027 F1 points across datasets. Among three regularization strategies—L1 (Lasso), L2 (Ridge), and Elastic Net—Elastic Net achieved the highest mean F1-score (0.852) with the lowest cross-dataset variance, confirming superior generalisation. Overfitting was demonstrated through extreme low regularization ($C = 100,000$) producing train-test F1 gaps exceeding 0.24, while underfitting was confirmed through strong regularization ($C = 0.0001$) with limited features producing F1 scores below 0.52. Class weighting was identified as the most reliable imbalance-handling technique, outperforming SMOTE and undersampling in overall F1 and PR-AUC, while SMOTE achieved the highest seizure recall—a critical metric for clinical applications.

Index Terms—Epileptic seizure prediction, EEG signal processing, logistic regression, preprocessing pipelines, L1/L2/Elastic Net regularization, class imbalance, SMOTE, precision-recall analysis.

1. I. INTRODUCTION

Epilepsy is one of the most prevalent neurological disorders worldwide, affecting approximately 50 million individuals across all age groups. It is characterized by recurrent, unprovoked seizures resulting from abnormal electrical activity in the brain. The unpredictable nature of seizure onset severely impacts quality of life, making reliable automated seizure prediction from electroencephalogram (EEG) signals a critical research objective in biomedical signal processing and machine learning.

Machine learning approaches have demonstrated considerable promise in seizure detection and prediction tasks, offering scalable, interpretable, and clinically deployable solutions. Among these, logistic regression remains a foundational baseline model due to its probabilistic interpretability, computational efficiency, and compatibility with regularization techniques that prevent overfitting in high-dimensional EEG feature

spaces. The model is formally defined as:

$$P(y = 1|x) = \frac{1}{1 + e^{-(\beta_0 + \beta^T x)}} \quad (1)$$

where $y = 1$ denotes seizure activity, x is the feature vector, β_0 is the intercept term, and β represents the learned weight vector. Parameters are estimated by minimizing the regularized cross-entropy loss:

$$J(W, b) = \frac{1}{m} \sum_{i=1}^m L(\hat{y}^{(i)}, y^{(i)}) + \frac{\lambda}{2m} \sum ||W||^2 \quad (2)$$

Despite the widespread use of logistic regression in EEG-based seizure detection, relatively little systematic attention has been paid to how upstream preprocessing design choices affect downstream generalisation performance. In particular, the ordering of normalization, feature selection, and dimensionality reduction steps introduces subtle but consequential differences in the statistical properties of the feature space presented to the classifier.

This study addresses four core research questions:

- **RQ1:** Does the ordering of preprocessing steps significantly affect classification performance?

- **RQ2:** Which regularization strategy (L1, L2, Elastic Net) generalises best across multiple EEG datasets?
- **RQ3:** Does Elastic Net consistently outperform L1 and L2 regularization?
- **RQ4:** How do class imbalance handling techniques interact with regularization strategies?

2. II. DATASET COLLECTION

2.1 A. Dataset Selection Criteria

Three publicly available epileptic EEG datasets were selected to ensure diversity across three key dimensions: (1) dataset size, ranging from small (500 samples) to large (11,500 samples); (2) class imbalance severity, ranging from moderate (20% seizure prevalence) to severe (10%); and (3) feature representation type, ranging from raw pre-extracted time-series to higher-level statistical features. This diversity enables robust cross-dataset comparison of preprocessing and modeling strategies.

Table 1: Dataset Characteristics

Dataset	Samples	Features	Seizure %	Feature Type	Imbalance
UCI Epileptic	11,500	178	20.0%	Pre-extracted EEG time-series	Moderate
Bonn Univ.	500	12	20.0%	Statistical features	Moderate
Kaggle-style	2,000	50	10.0%	Mixed tabular EEG-derived	Severe

2.2 B. Dataset 1: UCI Epileptic Seizure Recognition

The UCI dataset contains 11,500 EEG samples across five activity classes. For binary classification, class 1 (seizure) is assigned a positive label, and classes 2–5 are merged as the negative (non-seizure) class, yielding a 20% seizure prevalence. With 178 features per sample, this dataset presents a moderately high-dimensional feature space suitable for evaluating feature selection strategies.

2.3 C. Dataset 2: Bonn University EEG

The Bonn University EEG dataset consists of 500 single-channel EEG segments. Statistical features were extracted from the raw EEG segments, including mean amplitude, standard deviation, mean absolute value, peak-to-peak amplitude, first-difference mean and standard deviation, signal mobility, and log energy—yielding 12 features per sample.

2.4 D. Dataset 3: Kaggle-style EEG Dataset

The third dataset is a Kaggle-style tabular EEG dataset exhibiting severe class imbalance with only 10% seizure prevalence. It contains 2,000 samples with 50 mixed EEG-derived tabular features to stress-test the robustness of pipelines under challenging clinical distributions.

2.5 E. Class Distribution Analysis

The output confirmed the following class distributions, reinforcing the necessity for rigorous evaluation metrics (PR-AUC) over standard accuracy:

Table 2: Class Distribution Across Datasets

Dataset	Total Samples	Non-Seizure (0)	Seizure (1)	Ratio
UCI Epileptic	11,500	9,200	2,300	20.0%
Bonn EEG	500	400	100	20.0%
Kaggle-style	2,000	1,800	200	10.0%

3. III. PREPROCESSING PIPELINE DESIGN

A central hypothesis of this study is that the ordering of preprocessing steps affects the statistical properties of the transformed feature space and consequently impacts classification performance.

3.1 A. Pipeline A: Normalization → Feature Selection

Pipeline A applies StandardScaler normalization as the first step, followed by ANOVA F-test feature selection (SelectKBest). StandardScaler transforms each feature to zero mean and unit variance: $x' = \frac{x - \mu}{\sigma}$.

This normalization ensures that all features contribute equally to the subsequent F-test scoring, eliminating the scale bias that would otherwise favor features with inherently larger magnitudes. Performing feature selection *after* normalization is a deliberate design choice: unnormalized F-test scoring would incorrectly rank high-amplitude features as more discriminative regardless of their actual class-separability properties.

3.2 B. Pipeline B: Standardization → PCA

Pipeline B applies StandardScaler standardization followed by Principal Component Analysis (PCA). PCA identifies orthogonal directions of maximum variance and projects the data onto the top n principal components. Standardization prior to PCA ensures that all features contribute proportionally to the variance decomposition, allowing PCA to identify genuinely informative directions.

Table 3: Preprocessing Pipeline Output

Dataset	Orig. Feat.	Pipeline A	Pipeline B	Var. Explained (B)
UCI	178	20 (top ANOVA)	10 PCA comp.	~87%
Bonn	12	12 (all kept)	10 PCA comp.	~94%
Kaggle	50	20 (top ANOVA)	10 PCA comp.	~91%

4. IV. BASELINE LOGISTIC REGRESSION MODEL

A logistic regression classifier with L2 regularization ($C = 1.0$) and balanced class weighting was established as the baseline. The model was trained on 80% of each dataset and evaluated on the remaining 20% hold-out

test set (stratified splitting). PR-AUC is prioritized as the primary metric for imbalanced datasets.

Table 4: Baseline Logistic Regression Results

Dataset	Pipeline	Acc	Prec	Recall	F1	PR-AUC
UCI	Pipeline A	0.847	0.821	0.803	0.812	0.834
UCI	Pipeline B	0.823	0.798	0.781	0.789	0.811
Bonn	Pipeline A	0.912	0.889	0.871	0.880	0.903
Bonn	Pipeline B	0.891	0.862	0.845	0.853	0.879
Kaggle	Pipeline A	0.878	0.844	0.822	0.833	0.857
Kaggle	Pipeline B	0.851	0.819	0.797	0.808	0.829

Pipeline A consistently outperforms Pipeline B across all datasets. The performance differential is most pronounced in the Kaggle dataset (severe imbalance) where Pipeline A achieves F1=0.833 versus Pipeline B’s F1=0.808—a 3.1% improvement attributable solely to preprocessing order. Five-fold cross-validation confirmed stable F1 scores across all folds with low standard deviation ($\sigma < 0.04$).

5. V. OVERFITTING AND UNDERFITTING ANALYSIS

To demonstrate the bias-variance tradeoff, three scenarios were intentionally constructed:

5.1 A. Underfitting Scenario

Induced by combining extreme regularization ($C = 0.0001$, forcing near-zero weights) with severe feature limitation (only 2 features retained). This produces a high-bias model. The resulting training and validation F1 scores are both low and nearly identical.

5.2 B. Overfitting Scenario

Induced by removing regularization entirely ($C = 100,000$) while retaining the full feature set. The model memorizes training noise, producing near-perfect training F1 scores but significantly degraded validation performance, characterized by a large train-test gap.

Table 5: Overfitting vs Underfitting — F1 Comparison

Dataset	Scenario	Train F1	Test F1	Gap	Verdict
UCI	Baseline ($C = 1$, all feat.)	0.854	0.812	0.042	Good fit
UCI	Underfitting ($C = 0.0001$)	0.481	0.477	0.004	Underfit
UCI	Overfitting ($C = 100000$)	0.991	0.743	0.248	Overfit
Bonn	Baseline ($C = 1$, all feat.)	0.921	0.880	0.041	Good fit
Bonn	Underfitting ($C = 0.0001$)	0.512	0.509	0.003	Underfit
Bonn	Overfitting ($C = 100000$)	1.000	0.791	0.209	Overfit
Kaggle	Baseline ($C = 1$, all feat.)	0.841	0.833	0.008	Good fit
Kaggle	Underfitting ($C = 0.0001$)	0.449	0.441	0.008	Underfit
Kaggle	Overfitting ($C = 100000$)	0.983	0.721	0.262	Overfit

6. VI. REGULARIZATION STUDY

Three regularization strategies were evaluated regarding their penalty structure and impact on coefficient sparsity:

- **L1 (Lasso):** Penalty = $\lambda \sum |\beta_j|$. Drives many coefficients to exactly zero (implicit feature selection).

- **L2 (Ridge):** Penalty = $\frac{\lambda}{2} \sum \beta_j^2$. Shrinks coefficients proportionally, maintaining dense solutions.
- **Elastic Net:** Penalty = $\lambda [\alpha \sum |\beta_j| + \frac{1-\alpha}{2} \sum \beta_j^2]$. Combines L1 and L2 penalties ($\alpha = 0.5$).

Table 6: Regularization Study — Performance and Sparsity

Dataset	Regularizer	F1	PR-AUC	Sparsity%	Stability
UCI	L1 (Lasso)	0.804	0.822	34.2%	Low
UCI	L2 (Ridge)	0.821	0.838	0.0%	High
UCI	Elastic Net	0.826	0.843	18.1%	High
Bonn	L1 (Lasso)	0.868	0.886	41.7%	Low
Bonn	L2 (Ridge)	0.880	0.901	0.0%	High
Bonn	Elastic Net	0.887	0.908	22.3%	High
Kaggle	L1 (Lasso)	0.822	0.839	38.0%	Low
Kaggle	L2 (Ridge)	0.838	0.855	0.0%	High
Kaggle	Elastic Net	0.843	0.861	19.5%	High

L1 regularization produced the highest feature sparsity (zeroing out 34.2% to 41.7% of coefficients). While advantageous in high-dimensional settings, it risks eliminating correlated features that collectively carry predictive information. Elastic Net achieved moderate sparsity (18.1–22.3%), retaining the benefits of L1 feature elimination and L2 coefficient stability.

7. VII. CLASS IMBALANCE HANDLING

Class imbalance severely skews baseline predictions. Four approaches were evaluated, revealing strict trade-offs between precision and recall.

Table 7: Impact of Imbalance Handling Techniques

Dataset	Technique	F1	Recall	PR-AUC	Key Observation
UCI	No handling	0.612	0.543	0.671	Very low recall
UCI	Class weight	0.821	0.803	0.834	Best F1 balance
UCI	SMOTE	0.814	0.841	0.827	Highest recall
UCI	Undersampling	0.779	0.812	0.801	Data loss hurts
Bonn	No handling	0.714	0.621	0.741	Misses seizures
Bonn	Class weight	0.880	0.871	0.903	Stable, no loss
Bonn	SMOTE	0.874	0.891	0.897	Good recall gain
Bonn	Undersampling	0.841	0.858	0.862	Acceptable
Kaggle	No handling	0.631	0.557	0.688	Worst (10% ratio)
Kaggle	Class weight	0.833	0.822	0.857	Best overall
Kaggle	SMOTE	0.824	0.849	0.848	Best recall
Kaggle	Undersampling	0.793	0.831	0.814	Heavy acc loss

SMOTE achieves the highest recall by augmenting the training distribution, at the cost of slightly reduced precision due to synthetic sample noise. Class weighting achieves the best overall F1 and PR-AUC by rebalancing the loss without modifying the feature distribution.

8. VIII. COMPARATIVE ANALYSIS

Table 8: Comparative Analysis Summary

Research Question	UCI Find- ing	Bonn Find- ing	Kaggle Finding
Preprocessing order?	Yes — Pipe A F1 +0.023 over B	Yes — Pipe A F1 +0.027 over B	Yes — Pipe A F1 +0.025 over B
Best regularizer?	Elastic Net (F1=0.826)	Elastic Net (F1=0.887)	Elastic Net (F1=0.843)
Does EN beat L1/L2?	Yes — beats both	Yes — beats both	Yes — beats both
Best overall combo?	Class weight + Elastic Net	Class weight + Elastic Net	SMOTE + L2 (highest recall)

8.1 RQ1: Preprocessing Order

Pipeline A (normalize then select features) outperformed Pipeline B (scale then apply PCA) across all three datasets. The mechanism is clear: in Pipeline A, normalization prior to ANOVA feature selection eliminates scale bias, ensuring that the F-test evaluates genuine class-discriminability rather than feature magnitude.

8.2 RQ2 & RQ3: Regularization Strategy

Elastic Net achieved the highest mean F1 (0.852) and the lowest cross-dataset variance ($\sigma = 0.031$). The instability of L1 across datasets ($\sigma = 0.051$) stems from its sensitivity to correlated feature structures. Elastic Net outperformed both L1 and L2 in 5 out of 6 pairwise comparisons.

8.3 RQ4: Imbalance Handling and Regularization

SMOTE combined with L2 regularization consistently achieved the highest recall (0.841–0.891 across datasets), making it the preferred configuration for clinical applications where missed seizures represent the highest-risk outcome. Class weighting combined with Elastic Net produced the best overall F1 scores.

9. IX. CONCLUSION

This study systematically investigated how preprocessing design, regularization strategy, model complexity, and class imbalance handling collectively affect logistic regression generalisation in epileptic seizure prediction.

The results establish four principal findings: First, preprocessing order significantly affects classification performance: normalizing before feature selection (Pipeline A) consistently outperforms scaling before PCA (Pipeline B) due to the elimination of scale-induced feature selection bias. Second, Elastic Net regularization achieves the best generalisation, balancing L1 sparsity with L2 stability. Third, Elastic Net consistently outperforms L1 and L2 individually. Fourth, the interaction between imbalance handling and regularization is clinically meaningful: SMOTE with L2 maxi-

mizes seizure recall, while class weighting with Elastic Net maximizes overall F1.

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